NEOGUARD | ESP32-S3 SYSTEM CONNECTIONS

Every used GPIO is drawn to its peripheral. Module blocks represent the carrier cable interfaces.

MAX30102 / 0x57

J2

VCC 1 — V3V3
GND 2 — GND
SDA 3 — I2C_SDA_GPIO4 — 4 GPIO4_/_SDA
SCL 4 — I2C_SCL_GPIO5 — 5 GPIO5_/_SCL
Pin_5 5 — INT / unused

MLX90614 / 0x5A

J3

VCC 1 — V3V3
GND 2 — GND
SDA 3
SCL 4

DS1302 RTC

J4

VCC 1 — V3V3
GND 2 — GND
CLK 3 — RTC_CLK_GPIO6 — 6 GPIO6_/_CLK
DAT 4 — RTC_DAT_GPIO7 — 7 GPIO7_/_DAT
CE_/_RST 5 — RTC_CE_GPIO14 — 14 GPIO14_/_CE

EXTERNAL RGB LED

J7

RED_A 1 — LED_R_A — R3 330 — LED_R_GPIO16 — 16 GPIO16_/_RED
GREEN_A 2 — LED_G_A — R4 330 — LED_G_GPIO17 — 17 GPIO17_/_GREEN
BLUE_A 3 — LED_B_A — R5 330 — LED_B_GPIO18 — 18 GPIO18_/_BLUE
COMMON_K 4 — GND

330 ohms per colour; common cathode to GND.

DEV3V3 → 3V3 → 3V3_OUT

U1

ESP32-S3

DEVELOPMENT BOARD

Offboard controller
GPIO labels shown, not
physical header positions.

Native USB stays on
the development board.

Built-in Wi-Fi connects
to the HTTPS backend.

GND → GND

ST7789 TFT / 240 x 280

GPIO8_/_RST 8 — TFT_RST_GPIO8 — 5 RST
GPIO9_/_DC 9 — TFT_DC_GPIO9 — 6 DC
GPIO10_/_CS 10 — TFT_CS_GPIO10 — 7 CS
GPIO11_/_MOSI 11 — TFT_MOSI_GPIO11 — 4 MOSI
GPIO12_/_SCK 12 — TFT_SCK_GPIO12 — 3 SCK

J5

GND — 1 GND
V3V3 — 2 VCC
TFT_BL — 8 BL_/_BLK

SPI MICROSD MODULE

GPIO21_/_CS 21 — SD_CS_GPIO21 — 6 CS
GPIO40_/_SCK 40 — SD_SCK_GPIO40 — 3 SCK
GPIO41_/_MOSI 41 — SD_MOSI_GPIO41 — 4 MOSI
GPIO42_/_MISO 42 — SD_MISO_GPIO42 — 5 MISO

J6

SD_VCC — 1 VCC
GND — 2 GND

BUZZER + DRIVER

GPIO15_/_BUZZ 15 — BUZZ_GPIO15 — R6 1k — BUZZ_BASE

Q1 (3 / 2 / 1)

R7 10k — BUZZ_BASE → GND

D1 — BUZZ_VCC — J8 1 POSITIVE
BUZZ_SW — J8 2 NEGATIVE

Q1 PN2222A: E1 / B2 / C3
D1 1N5819: stripe/cathode toward buzzer +
Active-high driver; buzzer 20 mA maximum.

Power and carrier harness

File: NeoGuard-power.kicad_sch

Selected V3V3 powers the modules; it is chosen on sheet 2. DEV3V3 is the controller output, not a short to external 3V3.

Default screen is the working ST7789 TFT. A different OLED needs its own verified interface. GPIO40-42 must be available on the actual board.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /
File: NeoGuard.kicad_sch

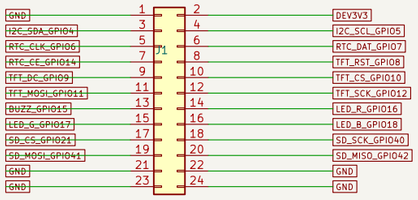
Title: NeoGuard – ESP32-S3 System Connections

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.4		Id: 1/2

NEOGUARD | POWER AND CARRIER HARNESS

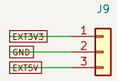
This sheet preserves the real carrier connector pin numbers and the selectable supply topology.

J1 / CONTROLLER CABLE



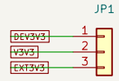
J1 is a cable connector, not a devboard socket.
Map each pin to the GPIO label on U1 / sheet 1.
Pin 2 is DEV3V3 from the ESP32 board.
Pins 1 and 21–24 are common ground returns.

INPUT SUPPLIES



J9 accepts already regulated supplies:
1 = external 3V3; 2 = GND; 3 = external 5V.
No onboard regulator or reverse-polarity protection.

JP1 / CARRIER 3V3 SOURCE



One shunt only:
1–2 = DEV3V3 -> selected V3V3
2–3 = EXT3V3 -> selected V3V3
Never fit both positions.

JP2 / SD MODULE SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the actual
SD breakout supply rating.

JP3 / BUZZER SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the buzzer
rating; this driver is for <=20 mA.

JP4 / TFT BACKLIGHT

Default OPEN / DNP.
Connect only for a verified 3V3-compatible
BL input with suitable driver/current limiting.

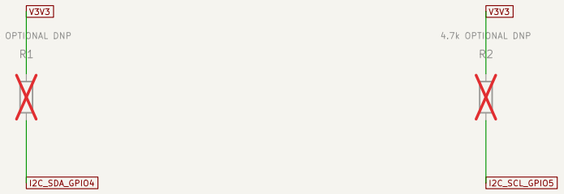
DECOUPLING / BULK STORAGE



Both copper layers use GND pours with thermals and stitching vias. Keep controller and modules powered together; signal isolation is not provided.

All carrier interfaces use 3.3V logic. Confirm breakout voltages, physical connector order, purchased-part pinouts and current budget before fabrication.

OPTIONAL I2C PULL-UPS



R1/R2 are DNP if module pull-ups already exist.
I2C pull-ups go to selected V3V3, never 5V.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /Power and carrier harness/
File: NeoGuard-power.kicad_sch

Title: NeoGuard – Power and Carrier Harness

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.4		Id: 2/2